



LIGHT -03

Class 10 - Science

1.
 - a. Draw a ray diagram to show the formation of image by a concave lens when an object is placed in front of it. [2]
 - b. In the same diagram mark the object-distance (u) and the image-distance (v) with their proper signs (+ve or -ve as per the new Cartesian sign convention) and state how these distances are related to the focal length (f) of the concave lens in this case.
 - c. Find the nature and power of a lens which forms a real and inverted image of magnification - 1 at a distance of 40 cm from its optical centre.*
2. An object 6 cm in size is placed at 50 cm in front of a convex lens of focal length 30 cm. At what distance from the lens should a screen be placed in order to obtain a sharp image of the object? Find the nature and size of the image. Also draw labelled ray diagram to show the image formation in this case. [2]
3. Define power of a lens. What is its unit? One student uses a lens of focal length 50 cm and another of -50 cm. What is the nature of the lens and its power used by each of them? [2]
4. Draw ray diagram showing the image formation by a convex lens when an object is placed at infinity. [2]
5. How is the refractive index of a medium related to the speed of light? Obtain an expression for refractive index of a medium with respect to another in terms of speed of light in these two media? [2]
6. If the image formed by a lens for all positions of an object placed in front of it is always erect and diminished, what is the nature of this lens? Draw a ray diagram to justify your answer. If the numerical value of the power of this lens is 10 D, what is its focal length in the Cartesian system? [2]
7. A ray of light travelling in air enters obliquely into water. Does the light ray bend towards the normal or away from the normal? Why? [2]
8. A convex lens of focal length 15 cm forms an image 10 cm from the lens. How far is the object placed from the lens? Draw the ray diagram. [2]
9. Draw a ray diagram to represent the nature, position and size of the image formed by a convex lens for the object placed at
 - i. infinity
 - ii. Between F_1 and optical centre (O) [2]
10. What is the power of a combination of lenses? [2]
11. A real image, $\frac{4}{5}$ size of the object is formed 18 cm from a lens. Calculate focal length of the lens. [2]
12. Draw ray diagram showing the image formation by a concave lens when an object is placed beyond twice the focal length of the lens. [2]
13. A student was asked to focus a permanent slide under high power of a microscope. Write the steps to be followed in a correct sequence. [2]
14.
 - a. If the image formed by a lens is diminished in size and erect, for all positions of the object what type of lens [2]

is it?

b. Name the point on the lens through which a ray of light passes undeviated.

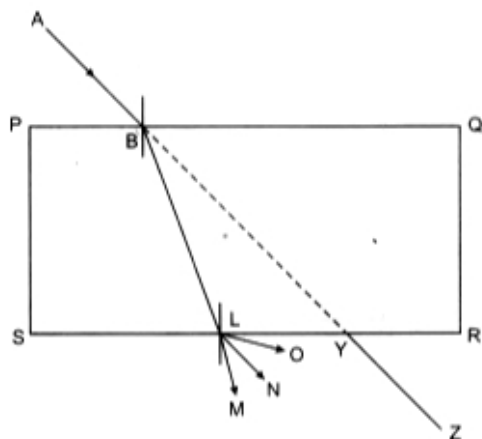
c. An object is placed perpendicular to the principal axis of a convex lens of focal length 20 cm. The distance of the object from lens is 30 cm. Find

i. the position,

ii. the magnification and

iii. the nature of the image formed.

15. If a light ray AB is incident on the surface PQ as shown, identify the correct emergent ray. [2]



16. For which position of the object does a convex lens form a virtual and erect image? [2]

17. Light enters from air to glass having refractive index 1.50. What is the speed of light in the glass? The speed of light in vacuum is $3 \times 10^8 \text{ ms}^{-1}$. [2]

18. One half of a convex lens is covered with a black paper. Will this lens produce a complete image of the object? Explain your observation. [2]

19. The power of a lens is +4D. Find the focal length of this lens. An object is placed at a distance of 50 cm from the optical centre of this lens. State the nature and magnification of the image formed by the lens and also draw a ray diagram to justify your answer. [2]

20. A concave lens made of a material of refractive index n_1 is kept in a medium of refractive index n_2 . A parallel beam of light is incident on the lens. Trace the path of rays of light parallel to principal axis incident on the concave lens after refraction when [2]

i. $n_1 > n_2$

ii. $n_1 = n_2$

Give reason for each.

21. Draw ray diagram showing the image formation by a concave lens when an object is placed at the focus of the lens. [2]

22. Draw a ray diagram showing the path of rays of light when it enters with oblique incidence [2]

i. from air into water,

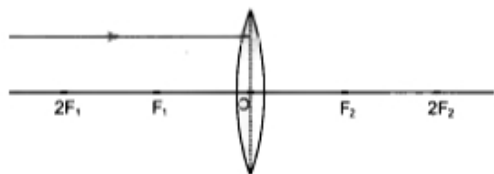
ii. from water into air.

23. A concave lens of focal length 15 cm forms an image 10 cm from the lens. How far is the object placed from the lens? Draw the ray diagram. [2]

24. Define 1 dioptre of power of a lens. [2]

25. An object is placed at a distance of 15 cm from a convex lens of focal length 20 cm. List four characteristics (nature, position, etc.) of the image formed by the lens. [2]

26. Draw the given diagram in your answer-book and complete it for the path of ray of light beyond the lens. [2]



27. Draw ray diagrams to show the formation of a three times magnified [2]

- real image,
- virtual image of an object kept in front of a converging lens. Mark the position of object, F, 2F, O and position of image clearly in the diagram.

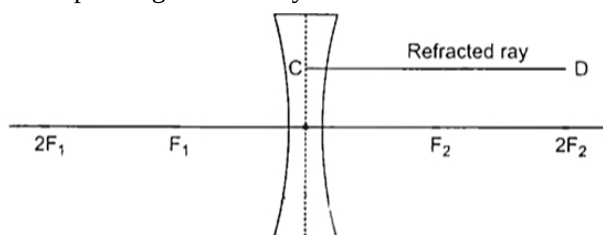
28. A candle is placed at the focus of a convex lens. What is the nature of beam of light produced by the lens? [2]

29. We wish to obtain an equal sized inverted image of a candle flame on a screen kept at distance of 4 m from the candle flame. [2]

- Name the type of lens that should be used.
- What should be the focal length of the lens and at what distance from the candle flame from the lens be placed.
- Draw a labelled diagram to show the image formation in this case.

30. Define power of a lens. The focal length of a lens is - 10 cm. Write the nature of the lens and find its power. If an object is placed at a distance of 20 cm from the optical center of this lens, according to the New Cartesian Sign Convention, what will be the sign of magnification in this case? [2]

31. The diagram below shows the refracted CD through a concave lens. Complete the diagram by drawing the corresponding incident ray. [2]



32. Find the power of a concave lens of focal length 2 m. [2]

33. A doctor has prescribed a corrective lens of power +1.5 D. Find the focal length of the lens. Is the prescribed lens diverging or converging? [2]

34. The power of a lens is +5 diopters. What is the nature and focal length of this lens? At what distance from this lens should an object be placed so as to get its inverted image of the same size? [2]

- Draw a ray diagram to show the refraction of light through a glass slab and mark an angle of refraction and the lateral shift suffered by the ray of light while passing through the slab. [2]
- If the refractive index of glass for light going from air to glass is $\frac{3}{2}$, find the refractive index of air for light going from glass to air.

36. With respect to air the refractive index of ice is 1.31 and that of rock salt is 1.54. Calculate the refractive index of rock salt with respect to ice? [2]

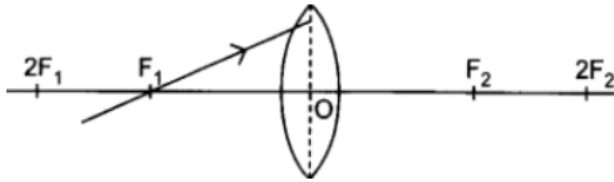
37. State and explain the factors on which lateral shift depends. [2]

38. The refractive indices of glass and water with respect to air are $\frac{3}{2}$ and $\frac{4}{3}$ respectively. If speed of light in glass is 2×10^8 m/s, find the speed of light in water. [2]

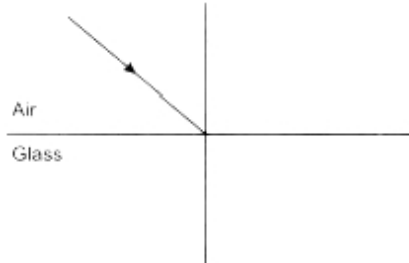
39. The refractive index of a medium x with respect to a medium y is $\frac{2}{3}$ and the refractive index of medium y with respect to medium z is $\frac{4}{3}$. Find the refractive index of medium z with respect to medium x. If the speed of light [2]

in medium x is $3 \times 10^8 \text{ ms}^{-1}$, calculate the speed of light in medium y.

40. Redraw the given diagram and show the path of the refracted ray. [2]



41. Complete the following diagram: [2]



42. Draw ray diagram showing the image formation by a convex lens when an object is placed between focus and twice the focal length of the lens. [2]

43. What are the laws of refraction of light ? How can these be verified experimentally ? [2]

44. Find the focal length of a lens of power -2.0 D . What type of lens is this? [2]

45. Draw ray diagram showing the image formation by a concave lens when an object is placed between focus and twice the focal length of the lens. [2]

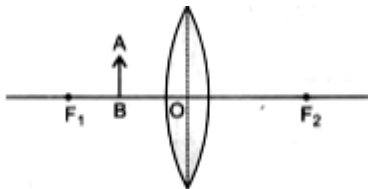
46. What is the relationship between the refractive index of two media? [2]

47. i. In refraction of light through a rectangular glass slab, the emergent ray is parallel to the direction of the incident ray. Why? [2]

ii. What happens when a light ray is incident normally on one of the faces of a rectangular glass slab?

48. a. The figure shows the position of an object AB in relation to a converging lens whose foci are F_1 and F_2 . [2]

Draw two rays to locate the position of image.



b. State three characteristics of image formed.

49. Distinguish between real and virtual image in a lens. [2]

50. The absolute refractive indices of glass and water are $\frac{4}{3}$ and $\frac{3}{2}$ respectively. If the speed of light in glass is $2 \times 10^8 \text{ m/s}$, calculate the speed of light in [2]

i. vacuum

ii. water.

51. i. An object of 5 cm height is placed at a distance of 20 cm from the optical centre of a concave lens of focal length 18 cm. Calculate (1) image distance and (2) the magnification in this case. [2]

ii. Compare the values of magnification obtained by a concave lens and a convex lens when both the lenses form virtual images.

52. In what S.I unit is the power of lens stated? A convex lens has a focal length of 50 cm. calculate its power? [2]

53. What happens after refraction, when: [2]

i. a ray of light parallel to the principal axis passes through a concave lens?

ii. a ray of light falls on a convex lens while passing through its principal focus?

iii. a ray of light passes through the optical centre of a convex lens?

54. The image of an object formed by a lens is of magnification -1. If the distance between the object and its image is 60 cm, what is the focal length of the lens? If the object is moved 20 cm towards the lens, where would the image be formed? State reason and also draw a ray diagram in support of your answer. [2]

55. A concave lens has focal length of 20 cm. At what distance from the lens a 5cm tall object be placed so that it forms an image at 15 cm from the lens? Also calculate the size of the image formed? [2]

56. Analyse the following observation table showing variation of image-distance (v) with object distance (u) in case of a convex lens and answer the questions that follow without doing any calculations: [2]

S. No.	Object distance u (cm)	Image distance v (cm)
1.	-100	+ 25
2.	-60	+ 30
3.	-40	+ 40
4.	-30	+ 60
5.	-25	+ 100
6.	-15	+ 120

i. What is the focal length of the convex lens? Give reason to justify your answer.

ii. Write the serial number of the observation which is not correct. On what basis have you arrived at this conclusion?

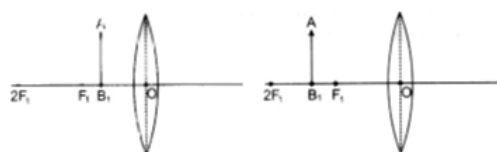
iii. Select an appropriate scale and draw a ray diagram for the observation at S.No. 2.

iv. Also find the approximate value of magnification.

57. Show diagrammatically, how should two converging lenses be arranged so that a parallel beam becomes parallel after passing through two lenses. [2]

58. Differentiate between a luminous and non-luminous source of light. [2]

59. Draw the following diagrams in your answer book and show the formation of the image of object AB by completing the ray diagrams. [2]



60.  [2]

i. What is the visible spectrum?

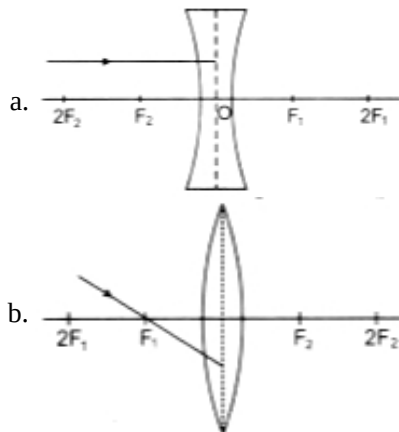
ii. Why is red used as the stopping light at traffic signals?

iii. Two triangular glass prisms are kept together connected through their rectangular side. A light beam is passed through one side of the combination. Will there be any dispersion? Justify your answer.

61. Velocity of light in diamond is $1.2 \times 10^8 \text{ ms}^{-1}$ and in vacuum, it is $3 \times 10^8 \text{ ms}^{-1}$, what is refractive index of [2]

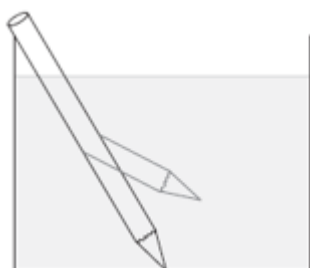
diamond ?

62. An object is kept at a distance of 1m from a lens of power +2D: [2]
i. Identify the type of lens.
ii. Calculate its focal length and distance of the image formed.
63. An object of height 10 cm is placed 25 cm away from the optical center of a converging lens of focal length 15 cm. Calculate the image-distance and height of the image formed. [2]
64. A convex lens can form a (i) real, inverted and magnified image as well as (ii) virtual, erect and magnified image [2] of an object. If the focal length of the lens is 10 cm, what should be the range of the object distance in both cases? Draw ray diagrams to justify your answer.
65. Redraw the diagram given below in your answer book and complete the path of ray. [2]



66. An object is placed at $2F_1$ in front of a convex lens. What is the [2]
i. Position
ii. Size
iii. nature of image?
67. a. The image of an object formed by a lens is of same size but inverted. If the object distance is 30 cm, [2]
calculate
i. The distance between the object and its image.
ii. Focal length of the lens.
b. Draw a ray diagram to show the image formed in above case.
68. A concave lens has a focal length of 10 cm. An object 2.5 cm high is placed 30 cm from the lens. Determine the [2]
position and size of the image.
69. State the two laws of refraction of light. [2]
70. Why does a light ray incident on a rectangular glass slab immersed in any medium emerges parallel to itself? [2]
Explain using a diagram.
71. Draw the path of a ray of light when it enters one of the faces of a glass slab at an angle of nearly 45° . Label on [2]
it (i) angle of refraction, (ii) angle of emergence and (iii) lateral displacement.
72. The image of a candle flame formed by a lens is obtained on a screen placed on the other side of the lens. If the [2]
image is three times the size of the flame and the distance between the lens and image is 80 cm, at what distance
should the candle be placed from the lens? What is the nature of the image at a distance of 80 cm and the lens?
73. Draw ray diagram showing the image formation by a convex lens when an object is placed at twice the focal [2]
length of the lens.

74.



- i. Which property of lightwave best described the above diagram?
- ii. The image formed looks slightly bent. Why?
- iii. On what factors the angle of refraction depends?

75. What is the power of lens ? What is its SI unit ?

[2]